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MFA600

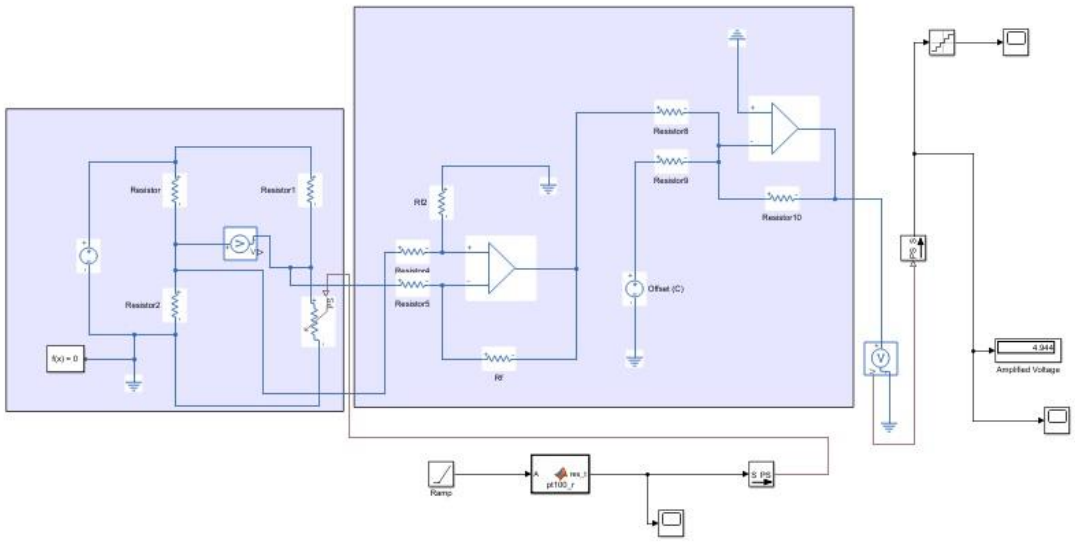
Project Report

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Shumail Alam Khan

AI and Automation
West University
Trollhättan, Sweden
shumail-alam.khan@student.hv.se

Abstract:
The current project proposes the design and simulation of a temperature data acquisition system (high precision) to be used in industry. In the system, a PT100 Resistance Temperature Detector (RTD) has been used, which is chosen because it is more accurate and stable in the required range of measurement which is 30°C to 119°C. This was accompanied by the creation of a complete signal conditioning circuit, including a Wheatstone bridge, with 3-wire compensation to remove lead resistance errors, and a precision instrumentation amplifier that was designed to scale the temperature range in the best way to a 5V ADC input. The most important design parameters were strictly computed, and they included: an amplifier gain of 6.079 and a -0.6717V offsets, a 20 Hz sampling rate, which was computed according to the Nyquist criteria of thermal processes, and the 16-bit ADC selection with 0.00169°C resolution. Analysis of the quantization errors ensured that maximum uncertainty was 0.00084°C showing outstanding accuracy of measurement. The entire signal path, including physical temperature to digital output, was numerically simulated and checked, giving a strong base of using reliable industrial temperature observation with emphasis on accuracy maximization at each stage of the acquisition pathway.



Question 1

Is the sensor you have chosen suitable for your temperature measurement range (justify your answer)?

Yes, a RTD (Resistance Temperature Detector) would be an appropriate sensor between the 30°C -119°C range. The one used in the project is PT100.

Justification:

- **Stability**: RTDs are very accurate and stable within the moderate temperature zone (0°C to 300°C) and the selected interval falls far within that range.
- **Sensitivity**: The RTD has a large resistance change per °C (sensitivity) which allows bridge voltages to be measured post-amplification by ADCs and this gives good ADC resolution.

- **Range**: RTDs have a large temperature range (can be used down to -200°C to up to 850°C) but are best used in moderate temperatures, such as 27°C to 113°C. They reach their limits best when they are not stretched.
- **Other Reasons**: They have a high degree of linearity, low drift and repeatability which is needed in laboratory level measurements. Its accuracy and reliability in data make it quite expensive compared to alternatives that are not expensive.

Question 2

Name another temperature sensor for your measurement range and state the pros and cons in relation to the primary sensor you have chosen.

Type K Thermocouple (TC) is other sensor which may be used instead of RTD. Some of them include the pros and cons as mentioned below.

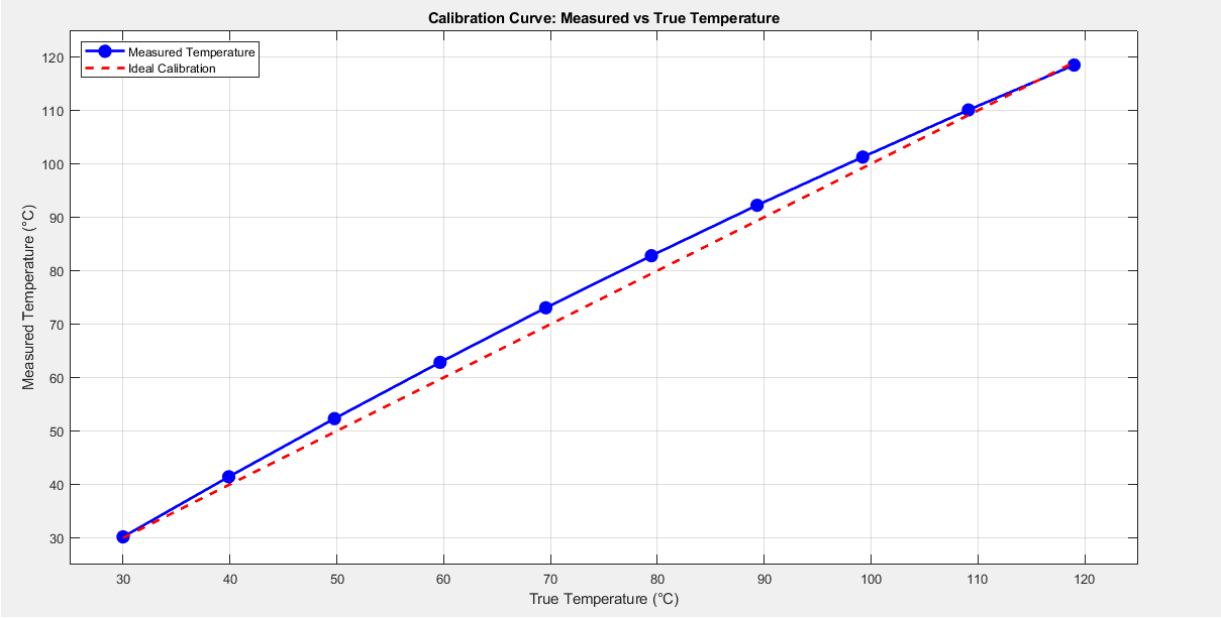
Pros:

- **Wider temperature range** (Type K typically −200 °C to +1260 °C) — more suitable for higher temperatures.
- **Very fast response** — decent for fast changing processes.
- **Rugged and inexpensive** — good for tough situations.

Cons:

- **Lower accurateness** and repeatability in the 30–119 °C range (typical uncertainties ±1–2 °C unless calibrated).
- **Requires cold-junction compensation (CJC)** in hardware or software to get absolute temperature, adding difficulty.
- **Greater drift** and less linearity at low-to-moderate temperatures than RTDs.

Conclusion: For 30–119 °C where accuracy and stability matter, Pt100 is preferable. Thermocouples are preferable where there is a need of speed and ruggedness or the temperature range is much larger.



Question 3

Does the sensor provide accurate temperature measurements? If not, what kind of calibration is required to compensate for the inaccuracies in the measurements?

The RTD as such is accurate within its given class (e.g. Class A). Measurement chain though creates errors. To make up these practical inaccuracies, calibration is needed.

Required Calibration:

- **Offset Calibration (Zeroing):** Find the system response at a known low temperature (e.g., 30 o C). The theoretical value should be equal to the output voltage. The deviation is fixable in the software or through a hardware trim. Amplifier offset voltage, input bias currents, gain error, noise and drift.
- **Gain Calibration (Spanning):** Hook up the system at a known high temperature (e.g., 1190C). Tune the system gain (in software or hardware) to attain the desired value of the output. Reference accuracy and quantization.

To obtain the greatest accuracy, a multi-point calibration (at 10+ temperatures as proposed) is to be carried out. The data may be used to construct a look up table or to fit a higher order polynomial in software which allows the system output voltage to be remapped to the actual temperature, and effectively makes the system linear.

Measuring amplifier (design + resistor choices)

Objective: Design an amplifier so that at T_max (119°C), V_out is close to 5V (using a 5V ADC reference).

Step 1: Determine RTD Resistance at Temperature Extremes

Using the IEC 60751 formula:

- At **30°C**: R30 ≈ 111.67 Ω
- At **119°C**: R119 ≈ 145.69 Ω

Step 2: Wheatstone Bridge Output Voltage

- **Bridge Supply Voltage, Vs=10V**
- **Fixed Resistors:** R1 = R2 = R3 = 100 Ω
  The differential output voltage Vo from the bridge is:

$$V_{bmin} = V_s \left(\left(\frac{R_2}{R_1 + R_2} \right) - \left(\frac{R_T}{R_{ref} + R_T} \right) \right)$$

- At **Tmin = 30°C**:

$$V_{bmin} = 10 \quad \left(\left(\frac{111.67}{111.67 + 100} \right) - (0.5) \right)$$

$$V_{bmin} \approx 0.275\ V$$

- At **Tmax = 119°C**:

$$V_{bmax} = 10 \left(\left(\frac{145.69}{145.69 + 100} \right) - (0.5) \right)$$

$$V_{bmin} \approx 0.933\ V$$

Step 3: Amplifier Gain Calculation using Slope Equation

We use the linear equation y = mx + c, where:

- y = Amplifier Output Voltage (V)
- x = Bridge Output Voltage (V)
- m = Amplifier Gain
- c = Amplifier Offset (V)

We have two known points the amplifier must satisfy:

1. At x1 = Vbmin = 0.275V, we want y1 = 1V
2. At x2 = Vbmax = 0.933V, we want y2=5V

This gives us the system of equations:

1 = m(0.275) + c (Eq.1)

5 = m(0.933) + c (Eq.2)

Subtracting Eq.1 from Eq.2 to eliminate c:

5 – 1 = m(0.933 – 0.275)

4 = m(0.658)

m = 4/0.6584

m ≈ 6.079

The gain mm is the required amplifier gain.

Required Gain ≈ 6.079

Now solve for the offset cc using Eq.1:

1 = (6.079)(0.275) + c

1 ≈ 1.6717 + c

C ≈ –0.6717 V

Step 4: Op-Amp Circuit and Resistor Selection
Using a standard non-inverting op-amp configuration:

$$\text{Gain} = 1 + Rg / Rf$$

$$6.079 = 1 + Rg / Rf \Longrightarrow Rg / Rf = 5.079$$

Choosing a standard value for Rg = 1.0 kΩ, we get:

$$Rf = 5.079 \times 1k\Omega = 5.079k\Omega$$

The closest standard E96 (1%) value is **5.11 kΩ**.

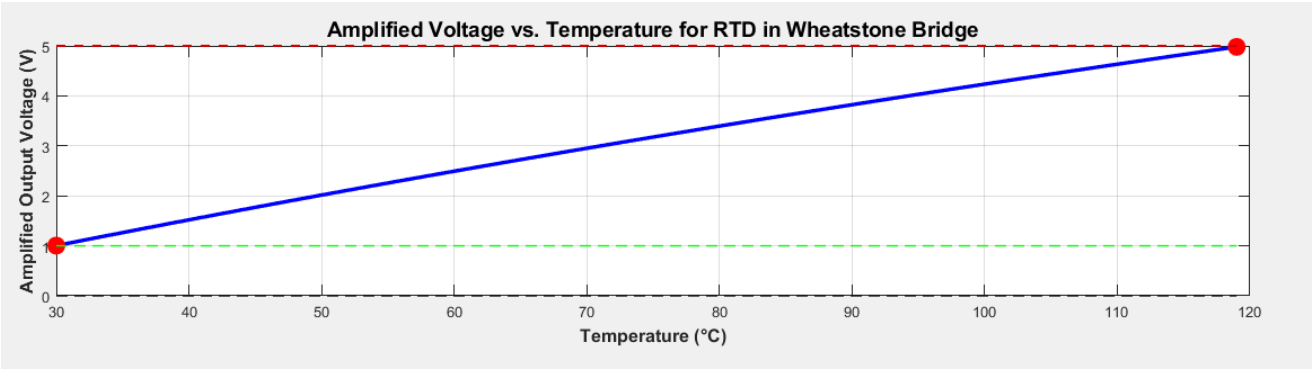
$$\text{Actual Gain} = 1 + 5110 / 1000 = 6.11$$

Step 5: Amplifier Output and Inaccuracies

- Vout at 119°C = (0.933V × 6.11) + (−0.6717V) ≈ 5.03V (Slightly over 5V, acceptable)
- Vout at 30°C = (0.275V × 6.11) + (−0.6717V) ≈ 1.01V

Inaccuracies from the Amplifier:

- **Resistor Tolerances:** 1% tolerance resistors can cause a gain error.
- **Input Offset Voltage:** Creates a constant DC error at the output, affecting the offset c.
- **Input Bias Currents:** Cause voltage drops across the resistors.
- **Temperature Drift:** The amplifier's parameters change with temperature.



Question 5

What is the minimum sample rate you should use and why? Now choose the number of bits in your AD converter (you choose yourself, justify your choice) and calculate what resolution you get with this number of bits. Add your calculation to the report.

The Nyquist-Shannon theorem is the minimum rate at which a signal can be reconstructed accurately, and it states that in order to reconstruct a signal accurately you need to sample at a rate at least twice the highest frequency component of the signal.

$$f_s \geq 2f_{max}$$

Where, fmax is the maximum frequency in the temperature signal. In the case of industrial temperature monitoring, thermal inertia is a limiting factor to signal bandwidth. Supposing the maximum temperature change rate to be 5 o C/s in rapid processes, and that in the interest of capturing the transient details a safety factor of 4x must be considered:

Thus, a sample rate of around 20 Hz is recommended to make sure that the rapid changes in temperature are accurately captured and a buffer band is provided over the Nyquist criterion.

ADC Bit Selection and Resolution
ADC Selection: 16-bit
Justification: A 16-bit ADC will give the high accuracy needed when measuring temperature where small variations (less than 0.1 o C) need to be identified. Having 65,536 discrete levels (216216), it is much more effective than lower-resolution ADCs but at the same time, it is economical to use in industry.

Resolution Calculation:
Voltage Resolution:

$$Qv = Vref / 2^n = 5 / 655365 \approx 76.29\mu V$$

System Sensitivity (from amplifier design):

- Voltage span: ΔV = 4.02 V (1.01V to 5.03V)
- Temperature span: ΔT = 89∘C (30°C to 119°C)

$$Sensitivity = \Delta T / \Delta V = 894.02 \approx 0.04517 \text{ V/}^\circ C$$

Temperature Resolution:

$$\Delta Tres = QV / Sensitivity = 76.29 \times 10^{-6} / 0.04517 \approx 0.00169^\circ C$$

The 16-bit ADC provides a temperature resolution of 0.00169°C, which is more than sufficient to resolve the Pt100's inherent accuracy of ±0.1°C.

Question 6

What will be your maximum error due to the quantization uncertainty (in degrees Celsius)? Suppose you have an AD converter with 5 V reference voltage and 10-bit resolution, what will be the resolution of the system? Compare it with 3.3 V reference voltage. Add your calculations. The calculations must show from physical value to digital value after ADC.

Maximum Quantization Error (16-bit ADC)
The maximum quantization error equals half of the ADC's resolution:
For our 16-bit ADC with 5V reference:

$$\text{Max Quantization Error} = QV / 2 = 76.29\mu V / 2 = 38.145\mu V$$

Convert to temperature error using system sensitivity (0.04517 V/°C):

Max Temperature Error = 38.145 × 10^−6 / 0.04517 ≈ 0.00084°C

The maximum quantization uncertainty is 0.00084°C - negligible compared to the Pt100's ±0.1°C accuracy.

Comparison of the models

ADC Configuration	Voltage Resolution	Temperature Resolution	Max Quantization Error
16-bit, 5V ref	76.29 µV	0.00169°C	**0.00084°C**
10-bit, 3.3V ref	3.22 mV	0.071°C	0.036°C
10-bit, 5V ref	4.88 mV	0.108°C	0.054°C

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